

HARSH GUPTA

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🐙 [Github](#)

Education

IIITDM Kancheepuram

Bachelor of Technology in Mechanical Engineering

2022 - 2026

Chennai, Tamil Nadu

Professional Experience

Software Developer Intern

June 2025 – December 2025

Unico International Pvt. Ltd., Bangalore

- Developed an **LLM-powered Task Management System** with automated task assignment, workflow orchestration, and real-time synchronization, designing end-to-end product workflows.
- Built and deployed a **Forex Trading Application**, built a scalable backend with real-time market data processing, trade execution pipelines, order management and P&L tracking using Supabase.

Technical Projects

Driver Drowsiness Detection | [Github Link](#)

- Built a **real-time driver drowsiness detection system** using **YOLOv8**, **OpenCV**, and **MediaPipe** to detect eye closure and yawning from live video streams.
- Trained **dual YOLOv8 models** on **53K+ images** and enhanced dataset quality using **GroundingDINO** auto-labeling and custom data pipelines.
- Integrated real-time inference with a **PyQt5 GUI**, confidence thresholding, and alert mechanisms for deployment-ready performance.
- Conducted model experimentation and validation using **TensorFlow/Keras** alongside PyTorch-based training workflows.

Brain Tumor Segmentation using U-Net | [Github Link](#)

- Developed a deep learning model using the **U-Net architecture** for automatic segmentation of brain tumors from MRI scans.
- Preprocessed data from the Kaggle LGG Brain MRI dataset with augmentation and normalization techniques.
- Achieved a **Dice coefficient of 0.83** and **IoU of 0.76** on the test set, demonstrating strong segmentation performance.
- Enhanced interpretability with **Grad-CAM** visualizations and deployed a demo via **Streamlit** for real-time usage.

Position of Responsibility

Member, CS Team

July 2023 – December 2024

Astra – Drone Racing Club, IIITDM Kancheepuram

- Trained a custom object detection model using **YOLO** on competition-specific shape datasets for autonomous mission tasks.
- Deployed the model on **NVIDIA Jetson Nano** for real-time onboard inference and UAV actuation.
- Integrated perception within a **ROS2** architecture to convert detections into navigation commands.
- Used **MAVROS** for MAVLink-based offboard control, enabling autonomous target approach and capture.
- Validated perception and navigation workflows in **Gazebo** before hardware deployment.

Technical Skills

Programming & Core Concepts: Python, C++, C, JavaScript, DSA, Data Analysis, Machine Learning, Deep Learning

Computer Vision : OpenCV, Image Processing, Object Detection(YOLO), Multi-Object Tracking(DeepSORT), Semantic Segmentation, Pytorch, Tensorflow, CNNs, Transformers, MediaPipe, GStreamer

Android Development: MVVM Design Pattern, Room Database, SQLite, Retrofit/Volley

Tools and Technologies: Firebase, MySQL, GitHub, Flutter, Jupyter Notebook, Supabase, ONNX, TensorRT, FastAPI

Robotics and Perception: ROS2, Perception System, SLAM

Coursework: Computer Vision, Data Science, Programming in Modern C++ (OOPs)